

PADHAISHURU

GATE Preparation Platform

GATE ME

Mechanical Engineering

TREND-BASED MOCK PAPER

— 02 —

Duration	3 Hours
Total Questions	65
Maximum Marks	100
Sections	11 subjects

Difficulty Distribution

Easy: 19 questions (29%)

Moderate: 34 questions (52%)

Difficult: 12 questions (18%)

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
These mock papers are constructed using analysis of historical GATE
PYQ patterns.
Historical patterns do not guarantee future examination questions.

GENERAL INSTRUCTIONS

GATE ME — Trend-Based Mock Paper

1. Examination Pattern

The paper contains 65 questions carrying a total of 100 marks. The duration of the examination is 3 hours.

2. Question Types

Questions are of three types:

- (a) Multiple Choice Questions (MCQ): One correct answer out of four options.
 - (b) Multiple Select Questions (MSQ): One or more correct answers out of four options. Each correct answer carries full marks. There is no negative marking.
 - (c) Numerical Answer Type (NAT): A real number answer to be entered. No options are provided.
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3. Marking Scheme

- 1-mark questions: +1 for correct, -0.33 for incorrect (MCQ only)
 - 2-mark questions: +2 for correct, -0.66 for incorrect (MCQ only)
 - 3-mark questions: +3 for correct, -0.66 for incorrect (MCQ only)
 - NAT questions carry no negative marking.
 - MSQ questions carry no negative marking.
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4. How to Answer

- For MCQ: Select exactly one option.
 - For MSQ: Select one or more options. Partial credit is NOT awarded.
 - For NAT: Enter the numerical answer in the provided space. Round off as specified.
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5. Rough Work

All rough work should be done only in the space provided in this question paper. Do not use additional sheets.

6. Calculator

An on-screen virtual calculator is available for use during the examination. Physical calculators are NOT permitted.

7. Important Notes

- Do not open the seal of the Question Booklet until you are instructed to do so.
 - The Question Booklet contains multiple pages. Verify the page count.
 - Immediately report any discrepancy in the Question Booklet to the invigilator.
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Q1. 1M [-0.33]

[General Aptitude]

Antonym of 'prolific':

- (A) Scarce
- (B) Abundant
- (C) Creative
- (D) Unusual

Q2. 1M [-0.33]

[General Aptitude]

Area of circle radius r:

- (A) $2\pi r$
- (B) πr^2
- (C) πr
- (D) $4\pi r^2$

Q3. 1M [no -ve]

[Engineering Mathematics]

Derivative of $\sin(x)$:

Enter your answer here

Q4. 1M [-0.33]

[Strength of Materials]

Section modulus $Z =$:

- (A) I/y
- (B) I/G
- (C) M/Z
- (D) $\frac{1}{2} \pi r^3$

Q5. 1M [no -ve]

[Strength of Materials]

Stress = Force /:

Enter your answer here

Q6. 1M [no -ve]

[Fluid Mechanics]

Laminar flow has:

Enter your answer here

Q7. 1M [-0.33]

[Fluid Mechanics]

Bernoulli applies to:

- (A) Viscous
- (B) Inviscid
- (C) Compressible
- (D) All

Q8.

1M [-0.33]

[Manufacturing Processes]

Sand casting uses:

- (A) Permanent mold
- (B) Sand mold
- (C) Die
- (D) Investment

Q9.

1M [-0.33]

[Manufacturing Processes]

MIG welding uses:

- (A) Flux
- (B) Gas shield
- (C) Both
- (D) None

Q10.

1M [no -ve]

[Thermodynamics]

First law based on:

Enter your answer here

Q11.

1M [-0.33]

[Thermodynamics]

Entropy change for reversible process:

- (A) Positive
- (B) Negative
- (C) Zero
- (D) Infinity

Q12.

1M [no -ve]

*[Heat Transfer]*Stefan-Boltzmann: $q = ?$

Enter your answer here

Q13.

1M [-0.33]

*[Heat Transfer]*Fourier's law $q =$:

- (A) $-k \, dT/dx$
- (B) $k \, dT/dx$
- (C) $k \, T$
- (D) $-k \, T$

Q14.

1M [-0.33]

[Theory of Machines]

Gear ratio is:

- (A) Teeth ratio
- (B) Speed ratio
- (C) Torque ratio
- (D) Power ratio

Q15.

1M [-0.33]

[Theory of Machines]

Cam converts:

- (A) Rotary to rotary
- (B) Rotary to reciprocating
- (C) Recip to recip
- (D) None

Q16.

1M [-0.33]

[Engineering Mechanics]

Equilibrium requires:

- (A) $\sum F_x = 0$
- (B) $\sum F_y = 0$
- (C) Both
- (D) None

Q17.

1M [-0.33]

[Engineering Mechanics] $F = ma$ is:

- (A) Newton 1st
- (B) 2nd
- (C) 3rd
- (D) None

Q18.

1M [-0.33]

[Machine Design]

Key prevents:

- (A) Sliding
- (B) Rotation
- (C) Both
- (D) None

Q19.

1M [-0.33]

[Machine Design]

Shaft designed for:

- (A) Torsion
- (B) Bending
- (C) Combined
- (D) Axial

Q20.

1M [-0.33]

[Engineering Mechanics]

Equilibrium requires:

- (A) $\sum F_x = 0$
- (B) $\sum F_y = 0$
- (C) Both
- (D) None

Q21.

1M [-0.33]

[Manufacturing Processes]

MIG welding uses:

- (A) Flux
- (B) Gas shield
- (C) Both
- (D) None

Q22.

1M [-0.33]

[Strength of Materials]

Stress = Force /:

- (A) Length
- (B) Area
- (C) Volume
- (D) Mass

Q23.

1M [-0.33]

*[Heat Transfer]*Fourier's law $q =$:

- (A) $-k \, dT/dx$
- (B) $k \, dT/dx$
- (C) $k \, T$
- (D) $-k \, T$

Q24.

1M [no -ve]

[Engineering Mechanics] $F = ma$ is:

Enter your answer here

Q25.

1M [-0.33]

*[Heat Transfer]*Stefan-Boltzmann: $q = ?$

- (A) σT^4
- (B) $\epsilon \sigma T^4$
- (C) σT^4
- (D) $\epsilon \sigma T^4$

Q26.

1M [-0.33]

[Engineering Mechanics] $F = ma$ is:

- (A) Newton 1st
- (B) 2nd
- (C) 3rd
- (D) None

Q27.

1M [no -ve]

[Engineering Mechanics]

Equilibrium requires:

*Enter your answer here***Q28.**

1M [-0.33]

[Strength of Materials]

Stress = Force /:

- (A) Length
- (B) Area
- (C) Volume
- (D) Mass

Q29.

1M [-0.33]

[Strength of Materials]

Section modulus Z =:

- (A) I/y
- (B) I/G
- (C) M/Z
- (D) $\frac{1}{Z} \int y^2 dA$

Q30.

1M [no -ve]

[Strength of Materials]

Section modulus Z =:

*Enter your answer here***Q31.**

2M [no -ve]

[Machine Design]

Key prevents:

*Enter your answer here***Q32.**

2M [no -ve]

[Theory of Machines]

Gear ratio is:

*Enter your answer here***Q33.**

2M [-0.66]

[Thermodynamics]

First law based on:

- (A) Energy conservation
- (B) Entropy
- (C) Temperature
- (D) Pressure

Q34.

2M [no -ve]

*[Heat Transfer]*Stefan-Boltzmann: $q = ?$

*Enter your answer here***Q35.**

2M [no -ve]

[Engineering Mechanics]

Equilibrium requires:

*Enter your answer here***Q36.**

2M [no -ve]

*[Engineering Mathematics]*Derivative of $\sin(x)$:

*Enter your answer here***Q37.**

2M [-0.66]

[Theory of Machines]

Gear ratio is:

- (A) Teeth ratio
- (B) Speed ratio
- (C) Torque ratio
- (D) Power ratio

Q38.

2M [-0.66]

*[General Aptitude]*Area of circle radius r :

- (A) $2\pi r$
- (B) πr^2
- (C) πr
- (D) $4\pi r^2$

Q39.

2M [no -ve]

[Manufacturing Processes]

Sand casting uses:

*Enter your answer here***Q40.**

2M [-0.66]

[Manufacturing Processes]

Sand casting uses:

- (A) Permanent mold
- (B) Sand mold
- (C) Die
- (D) Investment

Q41.

2M [-0.66]

[Manufacturing Processes]

Sand casting uses:

- (A) Permanent mold
- (B) Sand mold
- (C) Die
- (D) Investment

Q42.

2M [-0.66]

[Manufacturing Processes]

Sand casting uses:

- (A) Permanent mold
- (B) Sand mold
- (C) Die
- (D) Investment

Q43.

2M [no -ve]

[Fluid Mechanics]

Laminar flow has:

Enter your answer here

Q44.

2M [no -ve]

[Engineering Mechanics] $F = ma$ is:

Enter your answer here

Q45.

2M [-0.66]

[IC Engine]

Otto cycle is:

- (A) Constant volume
- (B) Constant pressure
- (C) Constant temp
- (D) Adiabatic

Q46.

2M [-0.66]

[Engineering Mechanics] $F = ma$ is:

- (A) Newton 1st
- (B) 2nd
- (C) 3rd
- (D) None

Q47.

2M [-0.66]

*[General Aptitude]*Area of circle radius r :

- (A) $2\pi r$
- (B) πr^2
- (C) πr
- (D) $4\pi r^2$

Q48.

2M [-0.66]

*[Heat Transfer]*Stefan-Boltzmann: $q = ?$

- (A) $<5@$
- (B) $<5E\tilde{a}$
- (C) $<5E\tilde{a}0$
- (D) $<5E\tilde{a}@$

Q49.

2M [no -ve]

*[Heat Transfer]*Fourier's law $q = :$

Enter your answer here

Q50.

2M [-0.66]

[Machine Design]

Key prevents:

- (A) Sliding
- (B) Rotation
- (C) Both
- (D) None

Q51.

2M [no -ve]

[Strength of Materials]

Stress = Force /:

Enter your answer here

Q52.

2M [-0.66]

*[Engineering Mathematics]*Derivative of $\sin(x)$:

- (A) $\cos(x)$
- (B) $-\sin(x)$
- (C) $-\cos(x)$
- (D) $\tan(x)$

Q53.

2M [-0.66]

[Theory of Machines]

Cam converts:

- (A) Rotary to rotary
- (B) Rotary to reciprocating
- (C) Recip to recip
- (D) None

Q54.

2M [-0.66]

[Machine Design]

Key prevents:

- (A) Sliding
- (B) Rotation
- (C) Both
- (D) None

Q55.

2M [-0.66]

[IC Engine]

Otto cycle is:

- (A) Constant volume
- (B) Constant pressure
- (C) Constant temp
- (D) Adiabatic

Q56.

2M [no -ve]

[Manufacturing Processes]

Sand casting uses:

*Enter your answer here***Q57.**

2M [no -ve]

*[Heat Transfer]*Stefan-Boltzmann: $q = ?$

*Enter your answer here***Q58.**

2M [-0.66]

[Theory of Machines]

Gear ratio is:

- (A) Teeth ratio
- (B) Speed ratio
- (C) Torque ratio
- (D) Power ratio

Q59.

2M [-0.66]

[Manufacturing Processes]

Sand casting uses:

- (A) Permanent mold
- (B) Sand mold
- (C) Die
- (D) Investment

Q60.

2M [-0.66]

[Fluid Mechanics]

Laminar flow has:

- (A) Low Re
- (B) High Re
- (C) $Re=1$
- (D) $Re=0$

Q61.

2M [no -ve]

[Machine Design]

Key prevents:

- (A) Sliding
- (B) Rotation
- (C) Both
- (D) None

Q62.

2M [-0.66]

[Manufacturing Processes]

MIG welding uses:

- (A) Flux
- (B) Gas shield
- (C) Both
- (D) None

Q63.

2M [no -ve]

[Thermodynamics]

First law based on:

*Enter your answer here***Q64.**

2M [-0.66]

[Manufacturing Processes]

MIG welding uses:

- (A) Flux
- (B) Gas shield
- (C) Both
- (D) None

Q65.

2M [-0.66]

*[Heat Transfer]*Stefan-Boltzmann: $q = ?$

- (A) $5 \times 10^{-8} \text{ W/m}^2\text{K}^4$
- (B) $5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$
- (C) $5.67 \times 10^{-10} \text{ W/m}^2\text{K}^4$
- (D) $5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$